

Module 11: Genomics & Biotechnology — Study Questions

1. What is a restriction enzyme, and what does it do to DNA?
2. What is recombinant DNA? Give an example of why a scientist would want to create it.
3. In your own words, describe how PCR copies DNA. How is PCR similar to DNA replication inside your cells (Module 08)?
4. What is gel electrophoresis used for? Do small or large DNA fragments travel farther through the gel, and why?
5. What are STRs (Short Tandem Repeats)? How are they used to identify individuals in forensics or paternity tests?
6. What is a GMO? Briefly describe the steps to create a bacterium that produces human insulin.
7. The genetic code is "universal" — the same codons code for the same amino acids in bacteria, plants, and humans. Why is this fact essential for genetic engineering? (Think back to the central dogma from Module 07.)
8. What is CRISPR-Cas9? How is it different from a restriction enzyme?
9. What is gene therapy? Give one example of a disease it could potentially treat.
10. Name one benefit and one concern about genetically modified organisms (GMOs).
11. **Connecting the modules:** A scientist discovers a mutation (Module 07) in a gene that causes a genetic disorder inherited in an autosomal recessive pattern (Module 09). She wants to use CRISPR to fix the gene. Explain, step by step, how the concepts from Modules 07, 09, and 11 all connect in this scenario.

12. **Thinking critically:** Should scientists be allowed to edit genes in human embryos (changes that would be passed to future generations)? State your position and give at least one reason for and one reason against.